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University Institute of Engineering (UIE)

Subject:	**Computer Organization and Architecture**
Case Study Title:	**Cache Memory Performance Analysis**
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Scenario

A processor accesses memory 1,000 times. Of these, 850 accesses are cache hits and 150 are misses. The cache hit time is 2 ns and the additional miss penalty is 50 ns.

Problem Statement

Calculate cache effectiveness and explain the relationship between hit ratio and average access time.

Case Study Questions

Question 01 :- Define cache memory.?

Solution :- Cache memory is a small and very fast memory. Cache memory is located inside the CPU. This memory stores most used data and instructions because CPU access that data quickly.
Example - When a CPU repeatedly uses the same data it can store that data in cache memory instead of accessing RAM every time.

Question 02 :- Distinguish Cache Hits from Cache Misses.?

Solution :-

Cache Hit	Cache Miss
Data found in cache.	Data is not found in cache.
Faster access.	Slower access.
Improves performance.	Reduces performance.
Requires less time.	Requires additional time.
Data access quickly.	The CPU has to access RAM.

Question 03 :- Calculate Cache Hit Ratio.?

Solution :- Total memory accesses = 1000, Cache hits = 850, Cache misses = 150

$$\text{Hit Ratio} \Rightarrow \frac{\text{Cache hits}}{\text{Total accesses}}$$

$$\Rightarrow 850 / 1000$$

$$\Rightarrow 0.85 = 85\ \%$$

Question 04 :- Calculate Cache Miss Ratio.?

Solution :- Miss Ratio => $\frac{\text{Cache Misses}}{\text{Total acesses}}$

=> 150 / 1000
=> 0.15
=> 0.15 * 100 = 15%

Hit Ratio + Miss Ratio = 1
85% + 15% = 100%

Question 05 :- Why is Locality of Reference Important.?

Solution :- Locality of Reference is the tendency of a program to access the same data or nearby data repeatedly within a short period of time. It is important because cache memory stores frequently or recently accessed data.

There are two main types :-

1. Temporal Locality : If data is accessed once, it is likely to be accessed again soon.
   Example : A variable used repeatedly inside a loop.

2. Spatial Locality : If one memory location is accesses, nearby memory locations are likely to be accessed soon.
   Example : Accessing elements of an array one by one.

Assignment :-

Calculate Average Memory Access Time (AMAT) with AMAT = Hit time + (Miss rate × Miss penalty).

Question :- Calculate Average Memory Access Time (AMAT) .?

Solution :- Given :
Total memory accesses = 1000, Cache hits = 850, Cache misses = 150,
Cache hit time = 2ns, Additional miss penalty = 50 ns

Step 1: Calculate Miss Rate

Miss Rate = 150 / 1000

= 0.15

Step 2: Apply AMAT Formula

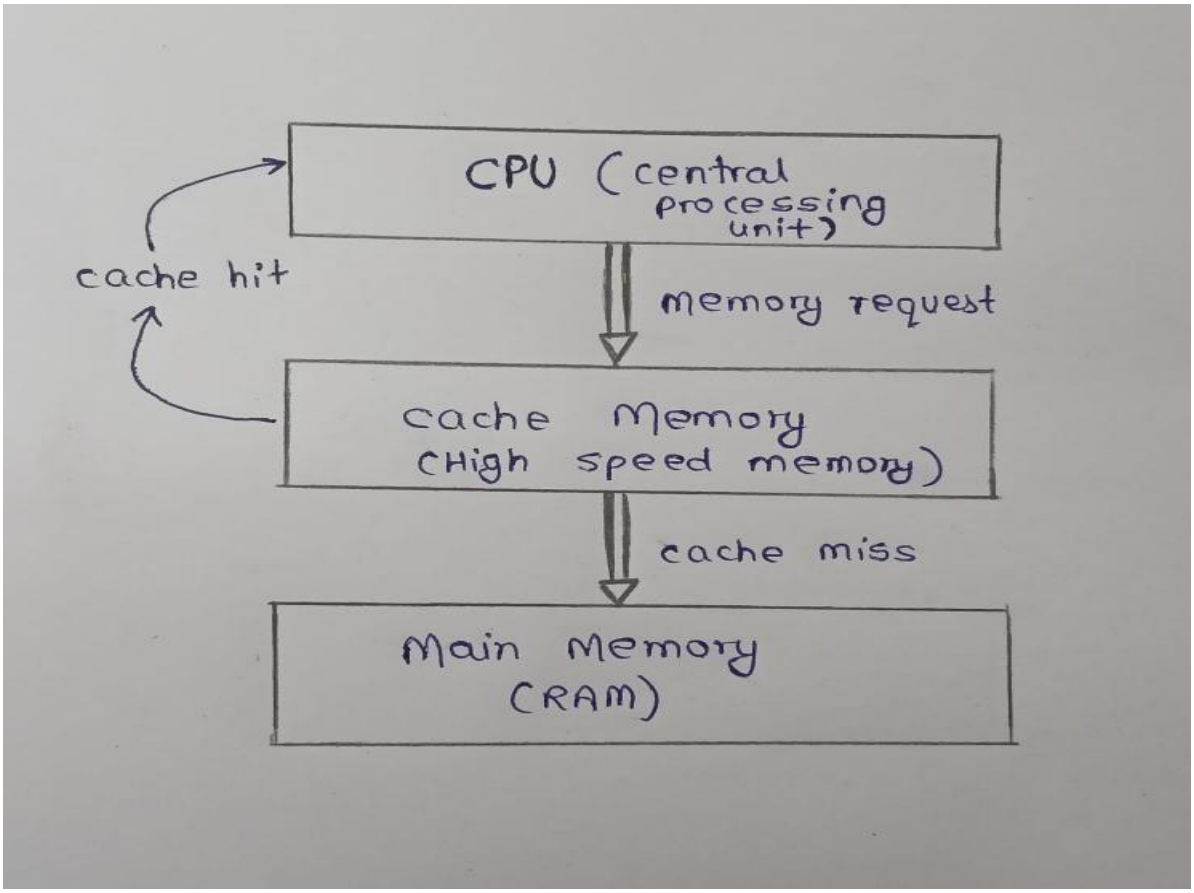
AMAT = Hit Time + (Miss Rate * Miss Penalty)

= 2 + (0.15 * 50)

= 2 + 7.5

= 9.5 ns

Question :- Draw a CPU–Cache–RAM diagram and list three observations.



Three Observations :

1. High Cache Hit Ratio: The cache hit ratio is 85%, which means most of the data requests are successfully found in cache memory.

2. Low Cache Miss Ratio: The cache miss ratio is 15%, which means only a few requests need to access the slower RAM.

3. Better System Performance: The Average Memory Access Time (AMAT) is 9.5 ns. A higher hit ratio reduces access time and improves CPU performance.